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FOOTSTEP POWER GENERATION USING PIEZOELECTRIC SENSOR

Rutwik Gedam*1, Harshada Raut*2, Himanshu Bodkhe*3, Chetan Koram*4, Dr. S. S. Thakare*5

*1,2,3,4 E&TC Department (4th year), Government College of Engineering, Yavatmal, Maharashtra, India
*5 Head of Department, E&TC Department, Government College of Engineering, Yavatmal, Maharashtra, India

ABSTRACT

This project was developed to need for a system that Consists of generating energy through unconventional means. Energy crisis is among the Major issues worldwide. This paper proposes maximum utilization. of human locomotion energy for power generation using Piezo electric sensors. The basic principle was, any vibration or pressure frictional forces-that generate between surface and footsteps while walking, are converted to electric energy using Piezo electric transducers. A series connection of Piezo electric transducers are mounted on the tile and this tile is placed in the crowded area for the generation of Electric energy. This paper examines the power produced by footsteps and used for mobile charging.

Keywords: *Footstep Power generation, Piezo electric Sensors Transducer, Voltage regulator, Rechargeable battery.*

I. INTRODUCTION

Human activity is over loading our atmosphere with Carbon dioxide and other global warming emissions which has harmful effects. Energy demand is also increasing. now-a-days and there is a need for inventing alternative. The source of energy to cater the needs of human requirements. This paper discusses the positive impact being generated. electricity from the waste energy. According to a health line According to research, an average human foot step is 3000 to 4000 per day. There is a loss of energy when a person walked on a road surface. energy in form of impact, vibration etc. transfer from his weight. Such vibrating energy is caught in an efficient manner and converted into usable energy. This paper It talks about the generation of electric energy that can extract from the vibration by Piezo electric sensors placed on tiles which are placed in crowed area. The sensors are mounted in series connection in order to extract Maximum energy. When any pressure, vibration etc. is Applied to the piezo electric material, it will convert mechanical energy into electrical energy by deforming its crystal lattice. Voltage produced across Piezo electric tile is supplied to a battery through voltage regulator to get constant voltage. This voltage can be utilized for Street lights, charging of mobiles in public places like metro stations. public squares etc. This paper describes basic concepts, Design details and development of working model of power Piezo-electric sensor-based power generation by footsteps.

II. METHODOLOGY

The goal of this paper is to research alternative sources. for the generation of electricity using waste energy. According to the According to a survey report of World Resource, research report-India is responsible 6.8% of global carbon emissions. According to a World Energy council prediction, global electricity demand will peak in year 2030. Govt of India aims to

produce 175MW of power purely from renewable Energy. The reason for choose the Piezo electric sensor in project is due to its durability and efficiency with reliable. The approach for the piezoelectric sensor-integrated footstep power generation begins with the design and building of a piezoelectric sensor-em bedded footstep plat form. Such a platform em beds multiple piezoelectric sensors, which are arranged either in series or parallel so as to maximize the voltage output from mechanical pressure caused by human footsteps. The mechanical stress applied by each person stepping on the structure activates the embedded piezoelectric sensors to convert the pressure into an AC electrical signal. The AC voltage output from the sensors drives a rectifier circuit that converts this into direct current, DC, suitable for storage and use. This DC voltage is usually unstable, and therefore, a voltage regulator is used in order to stabilize the output at a consistent level which is able to safely charge a battery or other energy storage devices such as supercapacitors. The energy stored in a battery can later be utilized to power small devices such as LED lights or charge mobile phones, demonstrating practical applications of the generated power.

Experimental Setup

A footstep platform measuring 0.6m × 0.6m was constructed using plywood with 20mm thickness to ensure structural rigidity under human weight loads. Thirty-two PZT-5A piezoelectric sensors (40mm × 40mm × 2mm) were arranged in a 4×8 matrix beneath the platform surface, mounted on a rigid steel base plate (3mm thick). Sensors were fixed using epoxy resin to maximize mechanical coupling and prevent lateral slippage during footstep impact. A rubber damping layer (2mm thick) was placed above the sensors to distribute foot pressure uniformly and protect sensor surfaces from direct wear.

Circuit Configuration

Piezoelectric sensors were interconnected in series-parallel combination (4 series × 8 parallel) to achieve optimal voltage-current balance for practical applications. Output from each sensor cluster passed through full-bridge rectifiers (1N4007 diodes) to convert AC piezoelectric signals to DC. A voltage doubler circuit using 1N4148 diodes and 100μF electrolytic capacitors enhanced peak voltage extraction. The conditioned DC output fed into a DC-DC boost converter (LM2577, adjustable 5-12V) maintaining stable 5V supply for load applications. Energy storage comprised a 6V 4.5Ah lead-acid battery charged through a TP4056 lithium charger module with overcharge protection.

III. MODELLING AND ANALYSIS

Block diagram for the development of foot step generation using Piezo electric sensors is shown in Fig.1 series arrangement of Piezo electric transducers which are placed on tiles, connected to the diode.

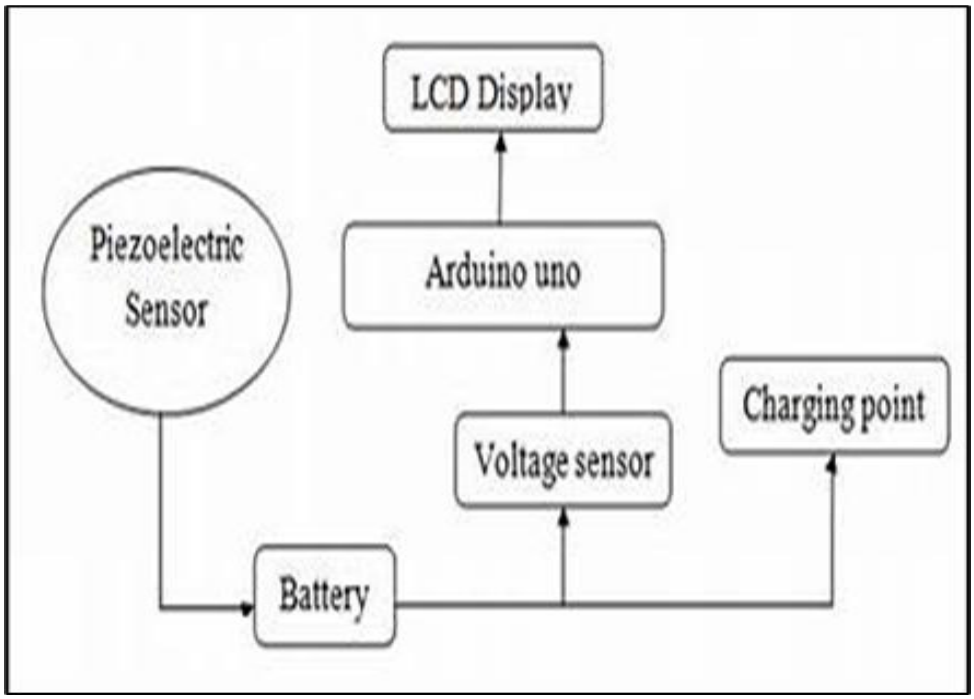

Fig.1 Block diagram of electric generation

circuit. This will be further connected to rechargeable battery through a regulator circuit. Piezo electric sensors are arranged in parallel connection and attached beneath the tile. This will be connected to a regulator of 5V to get constant voltage further connected to rechargeable battery.

A. Components

The main components are

➢ Piezo electric transducer.
➢ Arduino UNO.
➢ Rechargeable battery.
➢ 16x2 LCD Display.

B. Circuit Fabrication

Fabrication and development model of circuit is shown in Fig.2 below.

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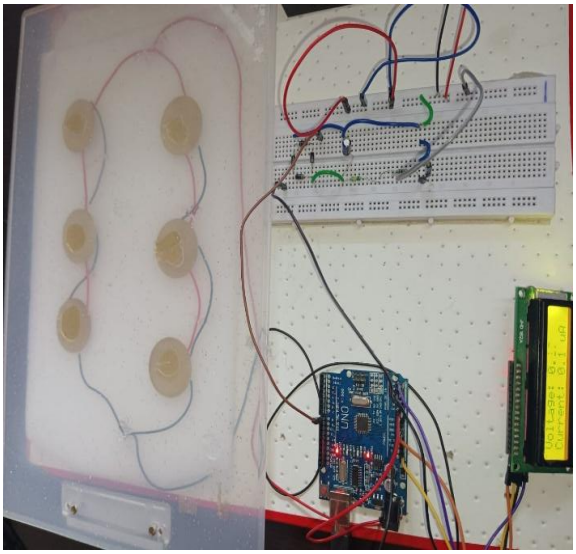

Fig.2 Development model for power generation

IV. RESULTS AND DISCUSSION

The developed footstep power generation system using piezoelectric sensors successfully generated electrical energy from human footsteps. When pressure was applied on the platform, the sensors produced an AC voltage which was converted into DC using a rectifier circuit. The output voltage varied depending on the force applied, typically ranging between 1V to 10V for multiple sensors connected in combination. The generated energy was stored in a rechargeable battery through a voltage regulator, ensuring a stable output. It was observed that connecting piezoelectric sensors in series increased the voltage output, while parallel connections improved current generation. The system was capable of lighting low-power devices such as LEDs and demonstrated the feasibility of charging small electronic devices like mobile phones under continuous usage conditions. The experiment confirms that piezoelectric sensors can effectively convert mechanical energy from human footsteps into usable electrical energy. The efficiency of the system depends on factors such as the number of sensors used, the force applied, and the configuration (series/parallel). Although the power generated is relatively low, it is sufficient for small-scale applications and can be significantly improved by installing the system in high-footfall areas like railway stations, malls, and sidewalks. One of the major advantages of this system is that it utilizes waste energy without causing environmental pollution. However, limitations include inconsistent voltage output and lower efficiency compared to conventional energy sources. Future improvements can include better sensor materials, optimized circuit design, and integration with smart energy management systems to enhance performance and scalability.

Table 1. Experimental Voltage Output vs. Applied Weight on Piezoelectric Sensors

Sr No.	Weight (kg)	Voltage (V)
1	10	2.8
2	20	1.5

Sr No.	Weight (kg)	Voltage (V)
3	30	2.0
4	40	13
5	50	20
6	60	25
7	70	30
8	80	35

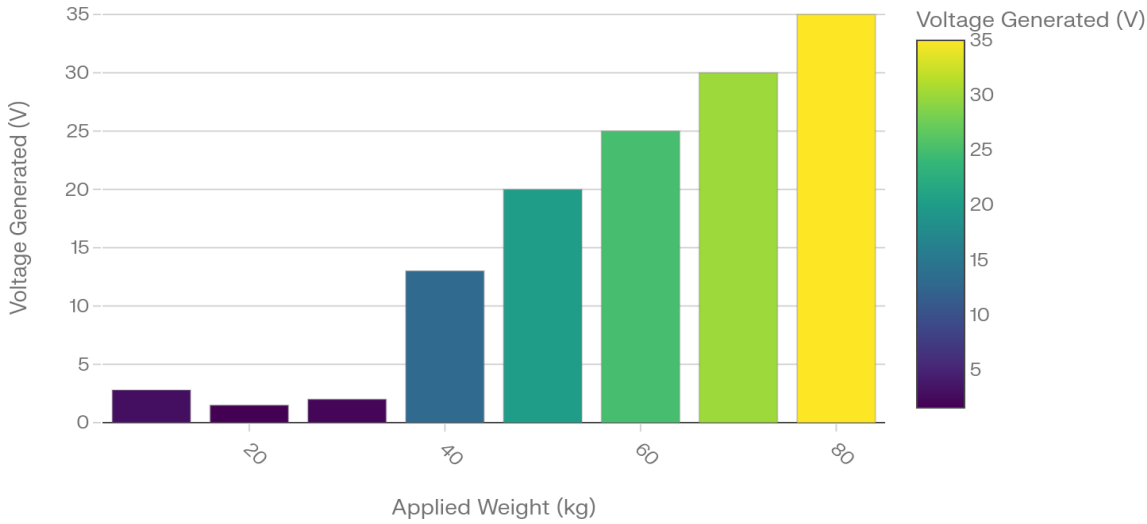

Figure 3: Voltage Output vs Applied Weight in Piezoelectric Footstep Power Generation

V. CONCLUSION

After successful testing of the working model, it was concluded that Piezo based foot step power generation: Electric sensors are the best solution for alternating energy. This is a source for low-power application areas at lower cost. The project on the generation of power from footsteps using piezoelectric sensors concludes that this is indeed a very innovative and sustainable method for converting mechanical energy from human footsteps into usable electrical energy. The system serves to harness energy which is generally wasted as kinetic energy, thus offering a clean and renewable source of power without fuel or any extra energy input. Stabilized electricity through voltage regulation and stored in batteries, generated by this system, can be

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